Adversarial Evasion of ML Intrusion Detectors in IoT Networks

Benchmark Study on CICIoT2023 | Portfolio Research Report

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EXECUTIVE SUMMARY & CORE RESEARCH QUESTIONS

RQ1 (Clean Baseline Detection):
How well do Random Forest, XGBoost, and a Multilayer Perceptron (MLP) detect IoT attack traffic?
- Random Forest and XGBoost achieve >99.0% test accuracy on 34-class multiclass and >99.2% on 8-family.
- However, macro F1 reveals severe minority-class degradation due to attack traffic volume skew:
  multiclass macro F1 is ~0.70 for trees and 0.59 for the MLP.

RQ2 (Adversarial Evasion & Transferability):
How much do accuracy and attack-class recall drop under gradient, constrained, and transfer attacks?
- White-box unconstrained PGD (eps=0.10) drops MLP accuracy from 91.51% to 41.92% on multiclass.
- Feature-constrained evasion (restricting perturbations strictly to sender-controlled size padding,
  traffic slowing, and flow duration extension while enforcing physical invariants):
  drops MLP multiclass accuracy from 91.51% to 42.65% (53.40% attack success rate).
- Transferability to Trees: Surrogate gradient attacks transfer with devastating effect:
  Random Forest multiclass accuracy drops from 99.01% to 29.16%, and XGBoost drops from 99.06% to 16.96%.

RQ3 (Defenses & Trade-offs):
Do adversarial training and feature squeezing restore robustness, and what is the clean-data cost?
- Adversarial Training (eps=0.02) successfully recovers robust accuracy from 41.92% to 72.81%
  under PGD, incurring a moderate clean-accuracy penalty of 11.10%.
- Feature Squeezing is ineffective on tabular network flow distributions: 4-bit squeezing reduces clean
  accuracy to 58.95% while only reaching 49.70% robust accuracy.

Figure 1: Baseline Clean Accuracy Across Detection Targets

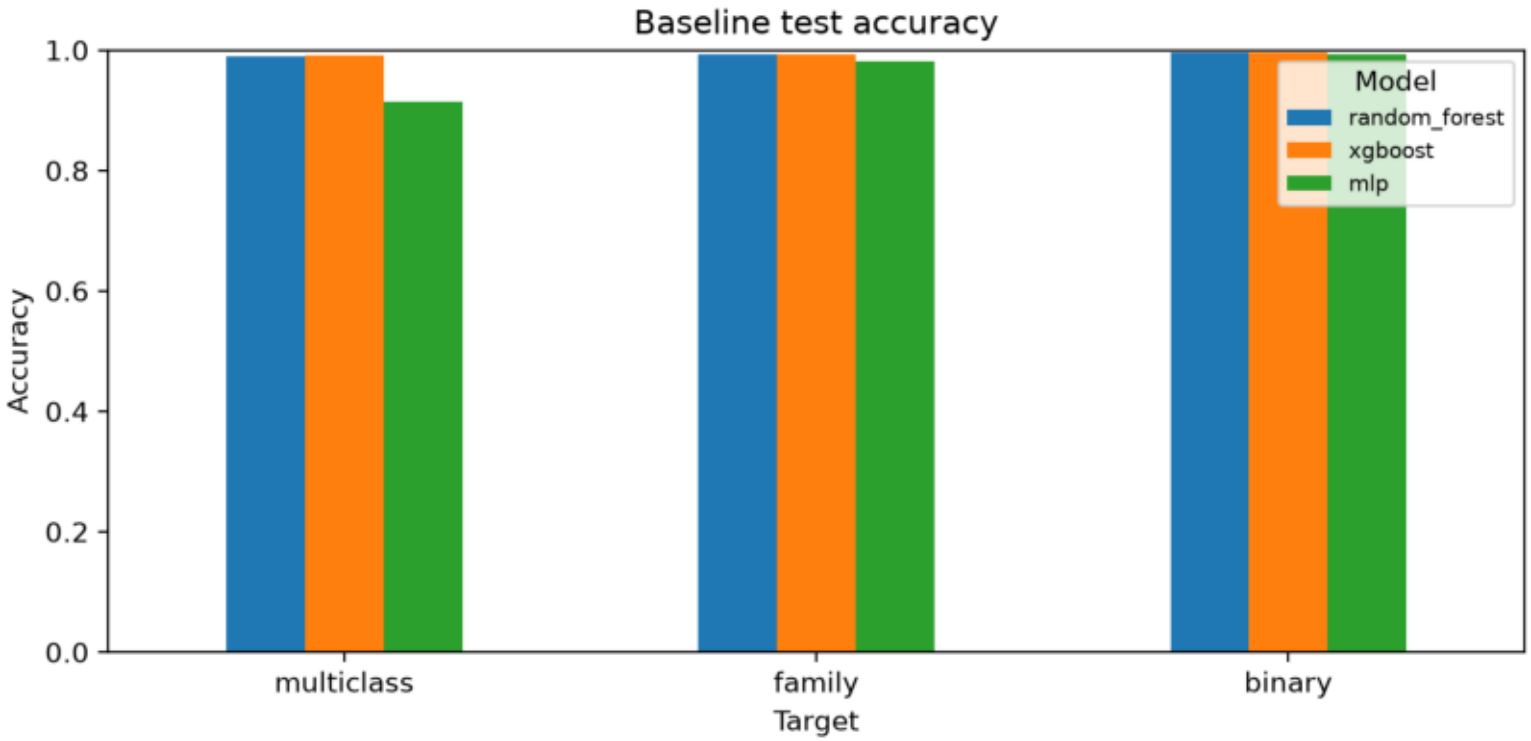


Figure 2: Multiclass Evasion Curves (MLP & Transfer to Trees)

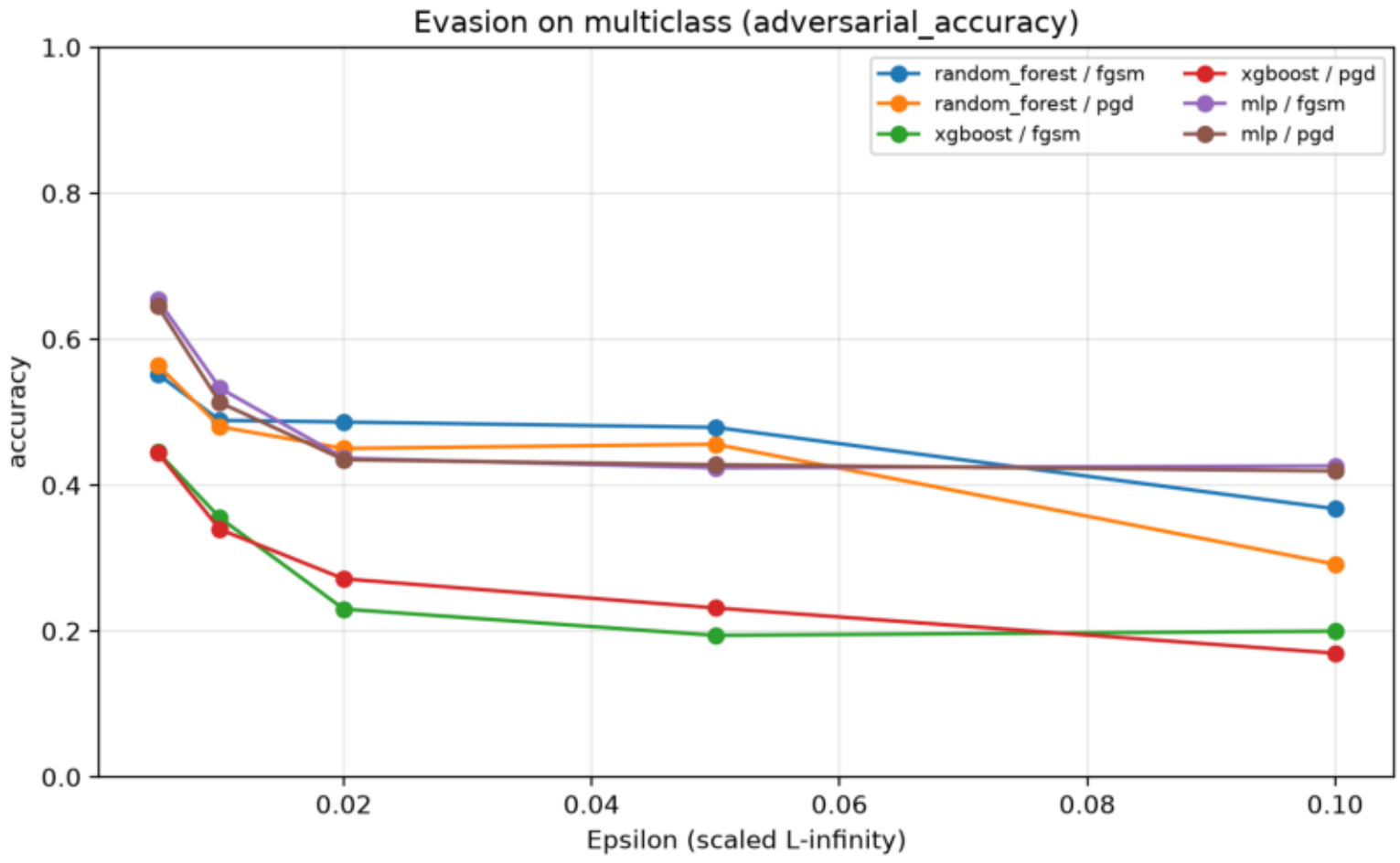


Figure 3: Defense Robustness vs Clean Accuracy Trade-off

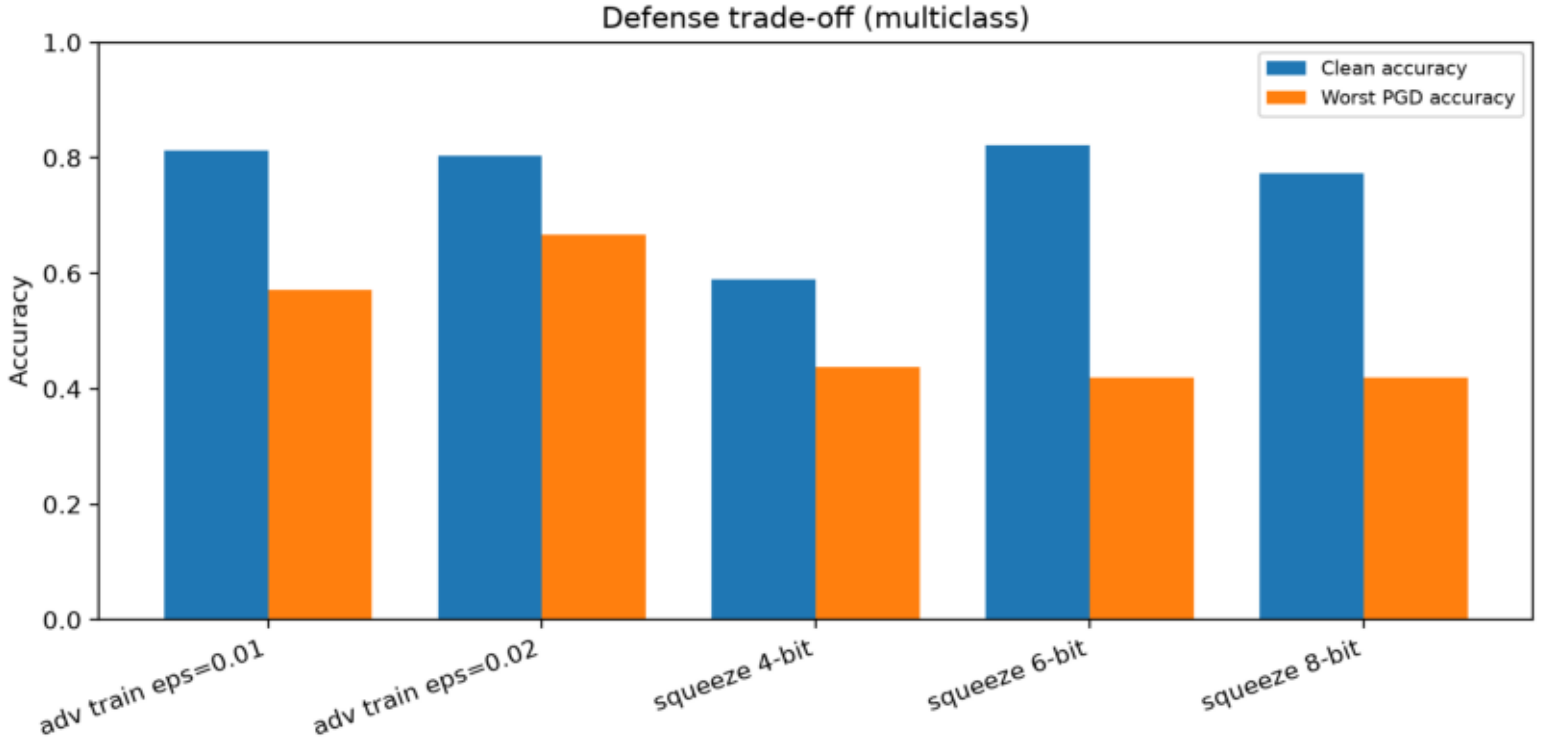


Figure 4: Per-Class Attack Recall Under Baseline Detectors

